

Spectral Theory and Variational Methods on Hilbert Spaces

This note surveys spectral theory and variational calculus on Hilbert spaces, and it deliberately exercises a broad range of mathematical typography so that an extraction pipeline can be evaluated against realistic notation. The text mixes ordinary prose with displayed equations in the same way that a typical research article does, alternating explanatory paragraphs and formal statements. Because the discussion is written in continuous sentences rather than tabular fragments, the body should be recovered as flowing text while the formulas contribute the special glyphs that motivate this fixture in the first place.

We begin with the geometry of an inner product space and the norm it induces. For vectors x and y in a Hilbert space, the inner product $\langle x, y \rangle$ is linear in its first argument and conjugate linear in its second, and the quantity below defines a genuine norm because it is positive, homogeneous, and satisfies the triangle inequality. These elementary facts underpin essentially everything that follows in the later sections.

$$\|x\| = \sqrt{\langle x, x \rangle} = \left(\sum_{j=1}^d |x_j|^2 \right)^{1/2}.$$

A direct consequence is the Cauchy and Schwarz inequality, which states that the magnitude of an inner product never exceeds the product of the corresponding norms, a bound that recurs throughout functional analysis and numerical estimation alike.

We now turn to linear operators and their spectra, the central objects of the theory. Let an operator be self adjoint, so that it equals its own adjoint, and recall that such operators have real spectra and admit a clean integral representation. The spectral theorem expresses the operator as an integral against a projection valued measure, and pairing that representation against a vector recovers a scalar spectral distribution that is often easier to analyze than the operator itself.

$$A = \int_{\sigma(A)} \lambda \, dE(\lambda), \quad \langle Ax, x \rangle = \int_{\sigma(A)} \lambda \, d\langle E(\lambda)x, x \rangle.$$

The resolvent exists and is bounded for every value outside the spectrum, and its norm grows as one approaches the spectrum, a behavior that quantifies how close a given number is to being an eigenvalue of the operator under study.

Variational methods provide a complementary viewpoint in which solutions arise as minimizers of an energy functional rather than as solutions of an equation. The functional below balances a gradient penalty against a source term, and its first variation vanishes precisely when the associated elliptic equation holds, which is the bridge between the analytic and the variational formulations of the same physical problem.

$$J[u] = \int_{\Omega} \frac{1}{2} \|\nabla u\|^2 - f u \, dx, \quad \partial_t J = 0 \Rightarrow -\nabla^2 u = f.$$

Here the gradient field $\vec{g} = \nabla u$ collects the partial derivatives of the solution, and floor and ceiling rounding, written $\lfloor t \rfloor \leq t \leq \lceil t \rceil$, appears when the continuous theory is discretized for computation on an actual machine.

For convenient reference we close with a compact inventory of the symbols used above, presented inline so that the surrounding discussion remains readable. The lowercase Greek letters $\alpha, \beta, \gamma, \delta, \epsilon, \theta, \kappa, \lambda, \mu, \nu, \xi, \pi, \rho, \sigma, \tau, \phi, \chi, \psi, \omega$ and the uppercase $\Gamma, \Delta, \Theta, \Lambda, \Xi, \Pi, \Sigma, \Phi, \Psi, \Omega$ name scalars and operators, while relations such as $p \approx q$, $u \sim v$, $s \equiv t$, $A \subseteq B$, $x \in S$, and the products $a \otimes b$ together with $a \odot b$ describe how those objects combine across the development.

$$\left\langle \prod_{i=1}^n (v_i \otimes w_i) \right\rangle = \left[\bigcup_k S_k \right] \cap \{z : |z| \leq 1\}.$$